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# GeoPops demonstration: An open-source, adaptable framework for agent-based modeling on synthetic populations

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# GeoPops

- Open-source package for generating geographically and demographically realistic synthetic populations for any US Census location using publicly available data
- Same methodology as GREASYPOP-CO (One Health Trust) but wrapped in convenient Python package that can be pip installed
- Automatic data downloading
- Compatibility with Census data beyond 2019 (in development)
- Compatibility with agent-based modeling software Starsim (IDM)

**github.com/ACCIDDA/GeoPops**

**pypi.org/project/geopops**

# MIDAS Notebook Tutorial

**Demonstration of framework**

1. Make a population of Howard County, MD
2. Run SIR Starsim model of generic upper respiratory infection
3. Incorporate prevalence-dependent "school closures"
4. Track outcomes by age and Census Block Group (CBG)

**github.com/ACCIDDA/GeoPops/tutorials/MIDAS**

# Output: Agents and networks

**Agents**

- Age
- Gender
- Race/ethnicity
- Worker status
- Student status
- School grade
- Commuter status
- Income category
- Workplace category
- Home CBG
- School CBG if applicable
- Work CBG if applicable
- GQ CBG if applicable

**School**

- List of students for each
- Size of grade levels
- Number of teachers
- CBG location

**Household**

- List of agents for each
- CBG location

**Workplace**

- Number of workers
- CBG if inside population area

**Group quarters (GQ)**

- Number of residents
- Number of staff
- CBG location

# Methods: Generating agents and households

**Public Use Microdata Samples (PUMS)**

**Combinatorial optimization (CO)**

**American Community Survey (ACS)**

- Subset of census survey responses
- Compositional details, e.g., the number of 5-member households in a CBG with school-aged children

- Demographic data at fine geographic scale
- Limited to marginal counts, e.g., the number of households in a CBG with 5 members and the number of households in a CBG with school-aged children

*Alexander Y. Tulchinsky, Fardad Haghpanah, Alisa Hamilton, Nodar Kipshidze, and Eili Y. Klein. (2024). Generating geographically and economically realistic large-scale synthetic contact networks: A general method using publicly available data. https://arxiv.org/abs/2406.14698*

# Methods: Assigning agents to schools and workplaces

| | Data source | Data | Methods |
|---|---|---|---|
| **Schools** | National Center for Education Statistics (NCES) | Student and staff counts for each grade level in public schools | • Students assigned to closest school that offered required grade<br>• Teachers assigned to school by selecting teachers from nearby CBGs |
| **Workplaces** | Longitudinal Employer-Household Dynamics (LEHD) | Origin-destination and workplace characteristics | • Iterative proportional fitting (IPF) to make origin-destination commute matrix<br>• IPF to make industry by destination commute matrix for each origin CBG<br>• Includes outside workers |

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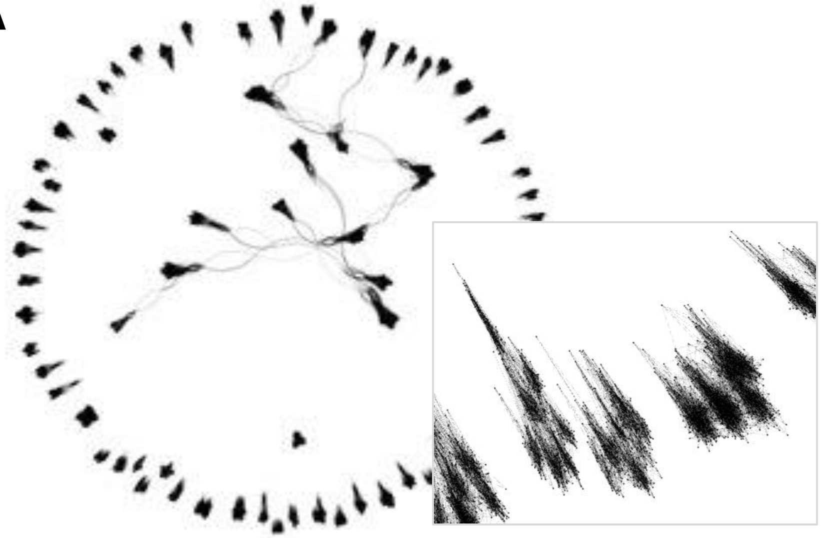
# Methods: Generating Contacts

| Layer | Algorithm | Parameters |
|---|---|---|
| **Household** | All agents within a household are fully connected | • N/A |
| **School** | Stochastic Block Model within each school | • Assortativity: $\alpha=0.9$<br>• Mean degree: $K=12$<br>• Blocks are grade levels (Pre-k – 12) |
| **Workplace** | Stochastic Block Model within each workplace | • Assortativity: $\alpha=0.9$<br>• Mean degree: $K=8$<br>• Blocks are income levels (<40k or ≥ 40k) |
| **Group quarters** | Watts-Strogatz Model within each GQ facility | • Mean degree: $K=12$<br>• Rewiring probability: $\beta=0.25$ |

*Alexander Y. Tulchinsky, Fardad Haghpanah, Alisa Hamilton, Nodar Kipshidze, and Eili Y. Klein. (2024). Generating geographically and economically realistic large-scale synthetic contact networks: A general method using publicly available data. https://arxiv.org/abs/2406.14698*

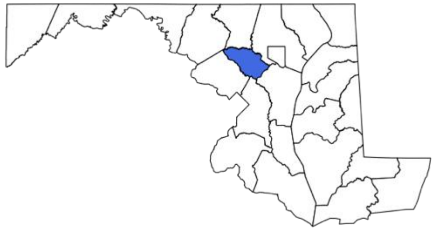
# Network visualizations for Howard County, MD

A



B

| p1 | p2 | beta |
|---|---|---|
| 748 | 78 | 1 |
| 984 | 883 | 1 |
| 1702 | 115 | 1 |
| 1709 | 1521 | 1 |
| 1875 | 651 | 1 |



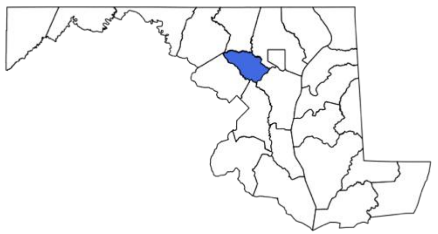
**Visualization and table snippet of the school network.** (A) Nodes are students and teachers clustered in grades within schools. (B) Same graph in Starsim tabular format where beta is the weight of the connection between person one and person two.

# Code example

```
import geopops

pars = {‘path’: ‘YOUR_OUTPUT_DIR’,    # Where you want outputs
       ‘census_api_key’: ‘YOUR_KEY’, # Your Census API key
       ‘main_year': 2019,            # Main year
       ‘geos': [‘24027’],            # State/county FIPS of main geo
       ‘commute_states’: [‘24’],     # Commute to/from states
       ‘use_pums’: [‘24’]}           # PUMS states to download

geopops.WriteConfig(**pars) # Set parameters
geopops.DownloadData()      # Download raw data files
geopops.ProcessData()       # Preprocessing for next steps
geopops.RunJulia()          # CO, network generation, export to csv
geopops.ForStarsim()        # Format for Starsim
```



# GeoPops with Starsim

**Components passed into Starsim to model disease transmission**

**SIR model**
- Respiratory disease
- Infection duration ~7 days
- Transmission coefficient ~0.1
- 50-day simulation

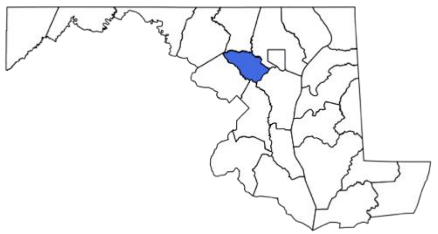
**GeoPops agents**

**Feedback module**
- Removes edges of school network when prevalence > 0.1

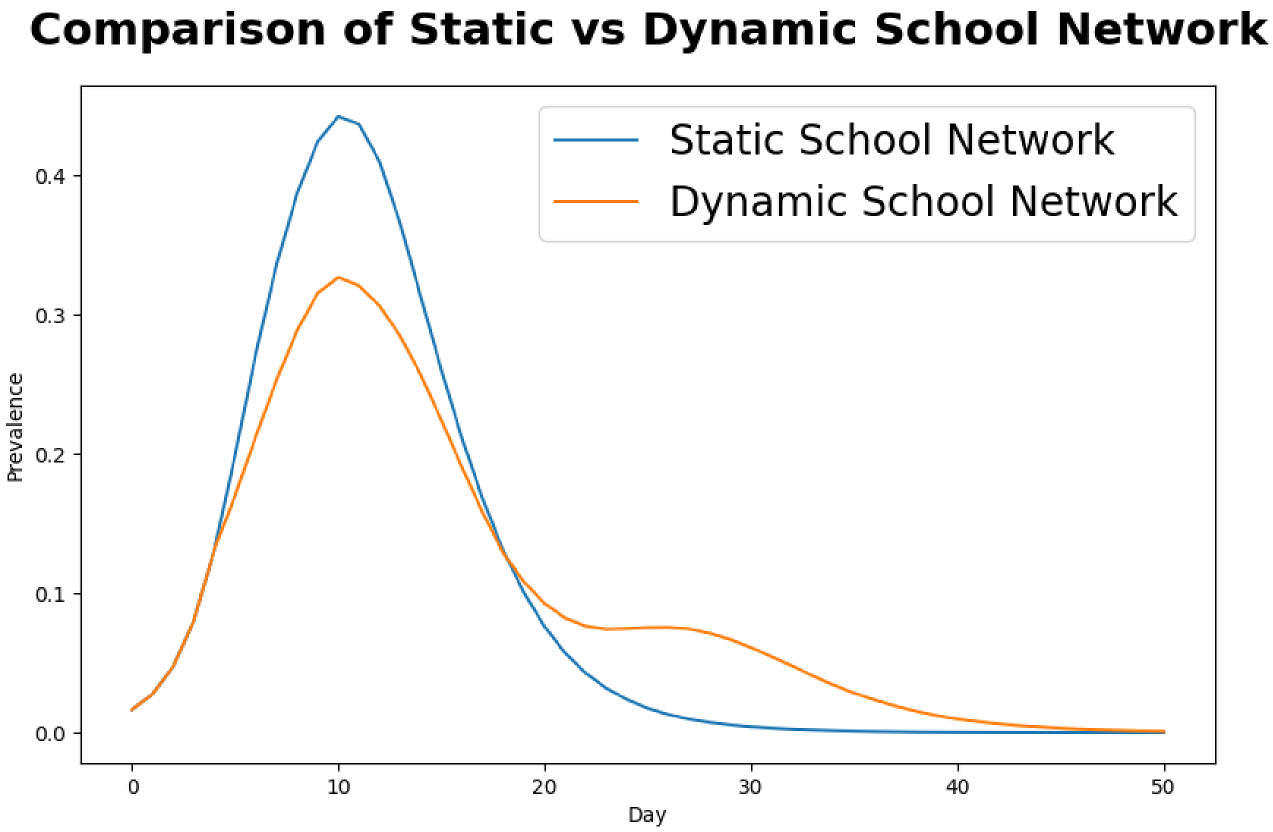
**GeoPops networks**

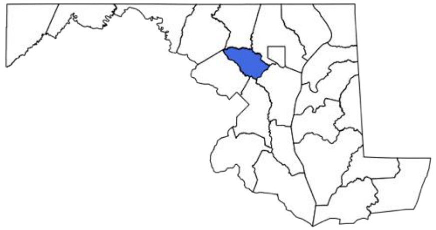
**Subgroup Tracking**
- Infections by age and CBG

**Simple example with generic upper respiratory infection for Howard County, MD**

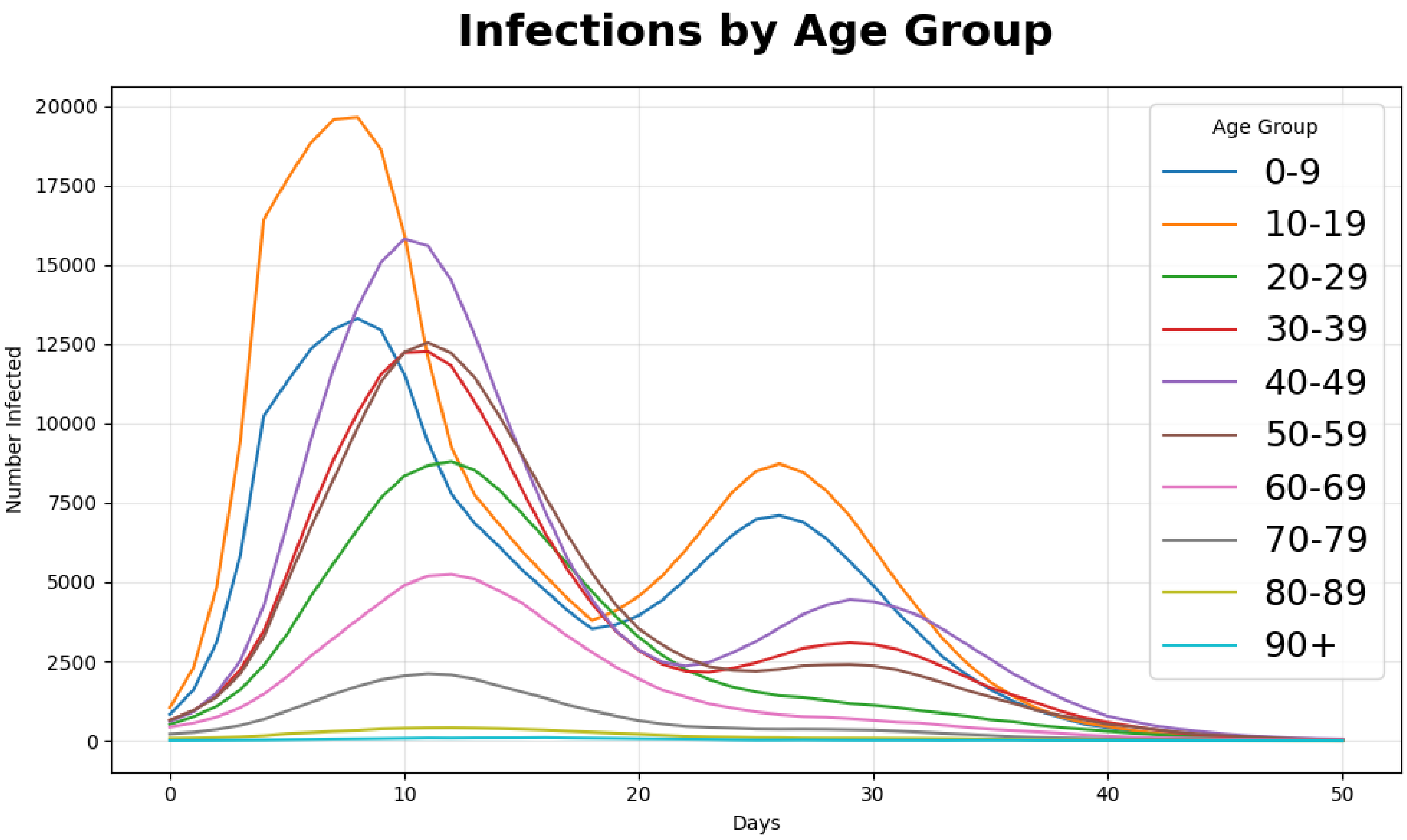


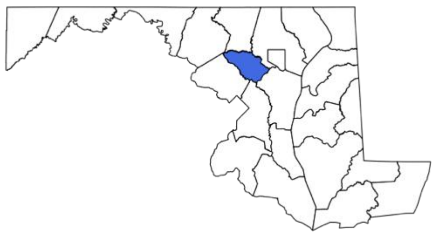
# Feedback vs No Feedback



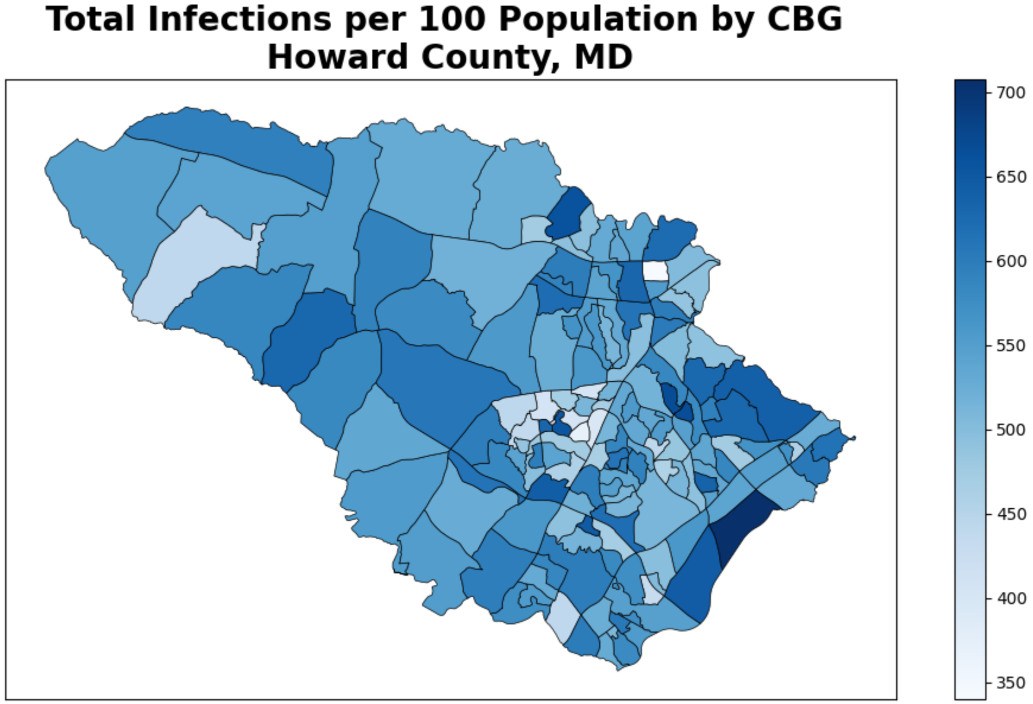



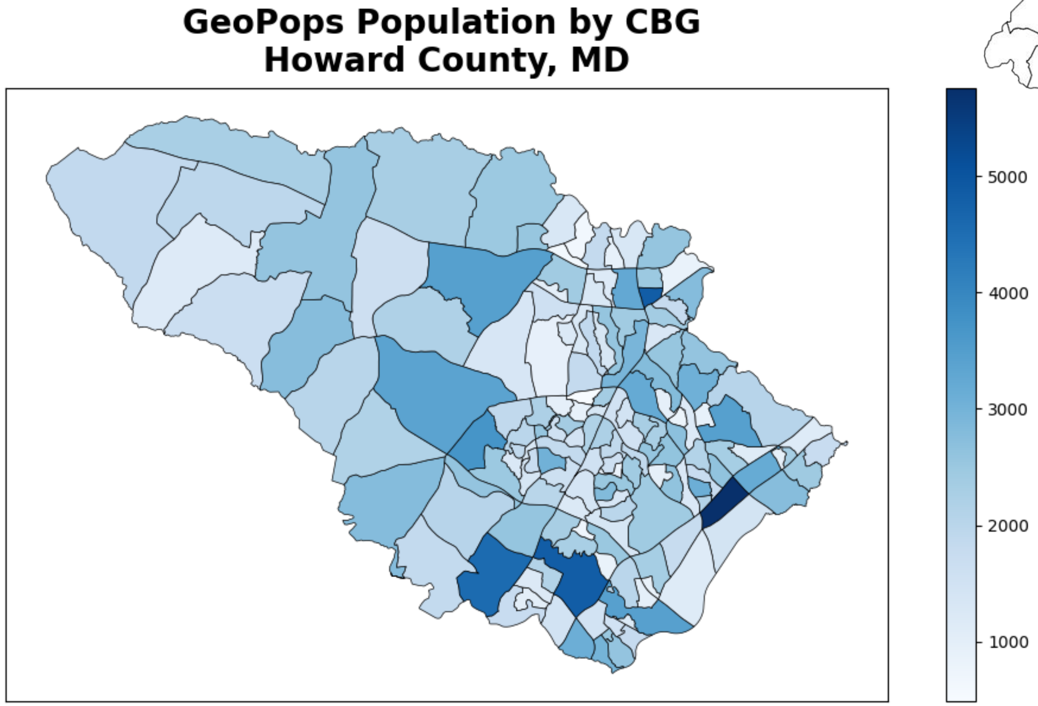
# Outcome tracking by demographic subgroup

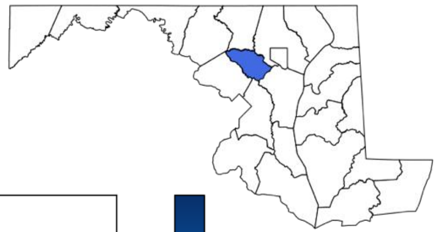


# Outcome tracking by geographic subgroup



# Why is this cool?!

**Open-source, user-friendly framework for capturing demographic and geographic differences in human behavior and health outcomes.**

- Reasonably realistic geographic and demographic granularity for night and day-time activities
- Don't have to build a model from scratch
- Can be used by state and county health departments for context-specific scenario modeling
- Seed infections to a specific geography or target interventions to specific demographic or geographic groups
- There are several research groups already using Starsim to model respiratory diseases that can now use GeoPops as well

# Conferences

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## Winter Simulation Conference 2025

**GeoPops: An open-source package for generating geographically realistic synthetic populations**

- More detailed methodology and usage
- Compare GeoPops to two other open-source synthetic population generators (UrbanPop and Geo-Synthetic-Pop)
- Make three MD pops (one with each generator)
- Compare individuals and homes to real Census data
- Compare network stats of each home, school, and workplace network
- Run Starsim COVID-19 model on each population
- Compare simulation results to each other and observed data for first wave of pandemic
- Discuss usability of each generator

## Epidemics 2025

**Modeling the impact of dynamic decision making on infectious disease outcomes by demographic and geographic subgroups: An open-source agent-based modeling framework**

- Run Starsim COVID-19 model with utility framework on GeoPops population of MD
- Individuals weigh health-wealth trade-offs of staying home from work as a function of their income and age
- Test different policy scenarios (e.g., paid sick leave, cash transfer) and compare to observed data from first wave of pandemic
- Demographic and geographic subgroup outcome tracking

# Questions

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- Pip install geopops and try going through the tutorial
- Log issues! Please be patient ☺
- Poster, slides, and Notebook tutorial on GitHub

**github.com/ACCIDDA/GeoPops/tutorials/MIDAS**

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